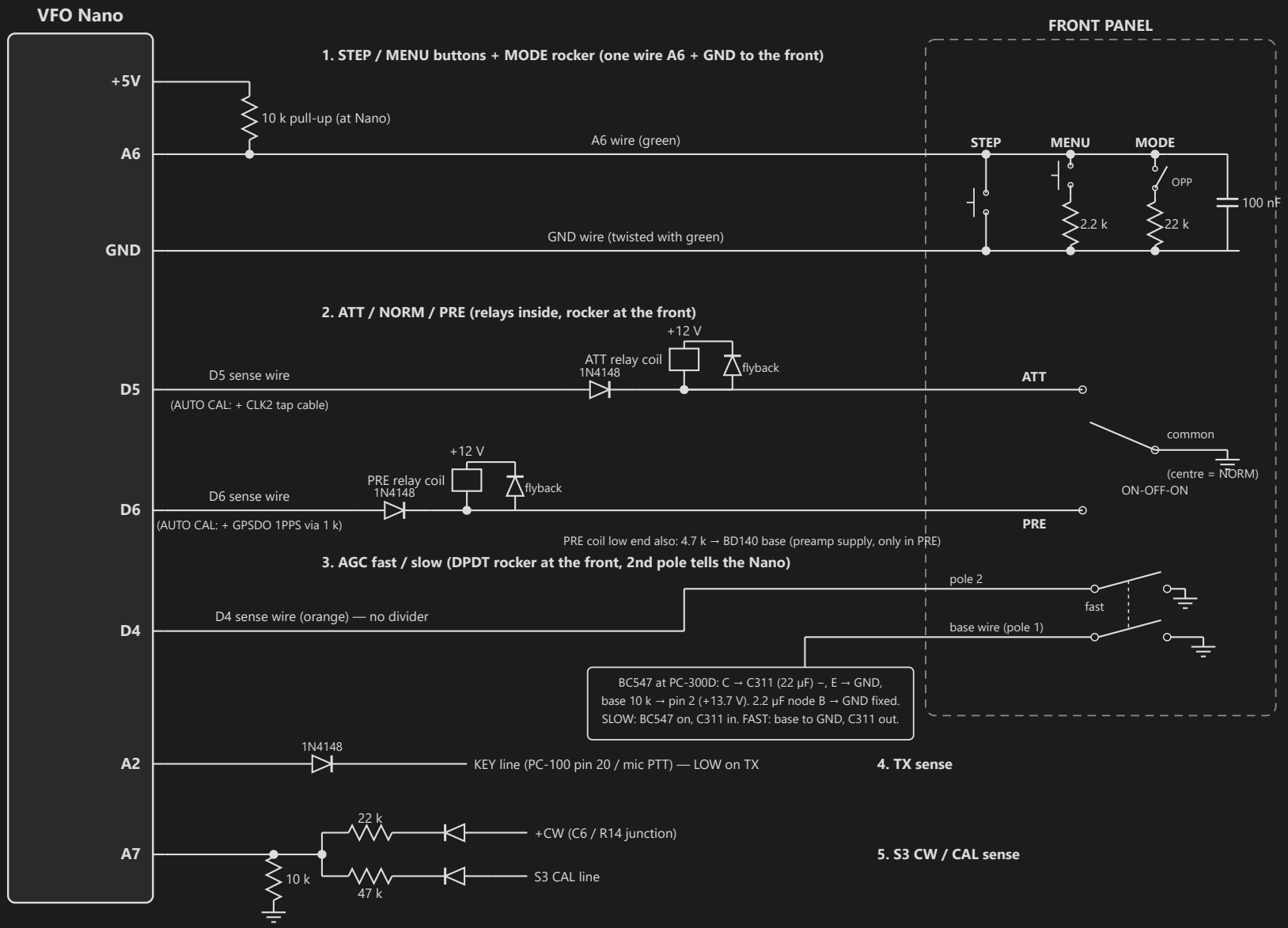
Atlas 210X — VFO Nano buttons, rockers, relays and sense wires

YT3CT · as built 2026-10-29 · diode band = cathode. Parts at the source end, plain wires to the Nano (5 V or less).



- From the Nano to the front / relays: **A6, GND, D4, D5, D6**. A6 readings: STEP 0, MENU ≈ 184 (170 with MODE), MODE alone ≈ 700, nothing 1023. A2 goes to the KEY line, A7 to the S3 network. The +12 V comes from the radio, never from the Nano.
- D5 / D6 / A2 / D4: Nano internal pull-up, LOW = on. For AUTO CAL (rocker on NOR) two removable cables join D5 (CLK2 tap from the CAL box, plain wire) and D6 (GPSDO 1PPS, 1 k at the D6 end); unplug them afterwards. Relay contacts (RF in / out of the π-pad and preamp) are not shown.
- A7: CW ≈ 3.8 V, CAL ≈ 2.1 V, REC / TRANS 0 V. CW_SENSE_FITTED 1 in the sketch (wired 2026-10-23).

Wires leaving the VFO box

Everything to run while the VFO box is open. Parts (diodes, resistors) at the far end unless noted.

#	Nano pin	Wire	Far end	Parts / notes
Rear socket J4 — twisted 4-wire set				
1	RST	grey	J4 pin 1	1 µF (105) in series, **at the Nano end** (10 nF is too small against the Nano's own 100 nF)
2	D1 (TX)	green	J4 pin 2	to adapter RX
3	D0 (RX)	white	J4 pin 3	from adapter TX
4	GND	black	J4 pin 4	chassis / Nano GND
Front panel buttons — twisted pair				
5	A6	green	STEP / MENU buttons, MODE rocker	10 k pull-up A6–5 V at the Nano; STEP 0 Ω, MENU 2.2 k, MODE (OPP) 22 k + 100 nF at the front
6	GND	twisted with green	buttons / MODE common	also the ATT/PRE rocker common
Sense inputs (Nano only listens, 5 V or less on the wire)				
7	D5	grey	ATT relay coil low end	1N4148, band to the coil. AUTO CAL: CLK2 tap cable joins here
8	D6	red	PRE relay coil low end	1N4148, band to the coil. AUTO CAL: GPSDO 1PPS cable (1 k at the D6 end) joins here
9	D4	orange	AGC rocker 2nd pole	DPDT: 2nd pole common = GND, “fast” throw = D4. No divider. Pole 1 (same wiring) grounds the BC547 base on PC-300D — never join the two poles
10	A2	white	KEY line	1N4148 fitted **at the Nano end** (band to the KEY line), well insulated — the white wire carries +13 V
11	A7	blue	S3 network	network → **orange** to CAL (1N4148 + 47 k), **white/black** to +CW (1N4148 + 22 k), 10 k to GND
Output / supply				
12	D13	white/red	VariBeam Nano D8 (through 10 k)	PTT request from CAT. VariBeam keys only after 300 ms HIGH (bootloader blinks ignored), drives D9 → BC547 → PTT relay (wiring sheet section 10)
RF				
13	Si5351 CLK2	thin coax	shielded CAL box → ATT/PRE J1	PC-820 removed. Box: 1 k, 2 × 10 Ω, 4.7 k, AUTO CAL tap header (wiring sheet section 9)

- Not from the VFO box: J4 pins 5–8 (audio via transformers, PTT to the KEY line), the rocker / relay +12 V wiring, the BD140 + 78L08 preamp supply and the AGC pole 1 base wire.
- Route along the chassis, cross coax at 90°. If a reading jumps on TX: 100 nF from that pin to GND at the Nano.